

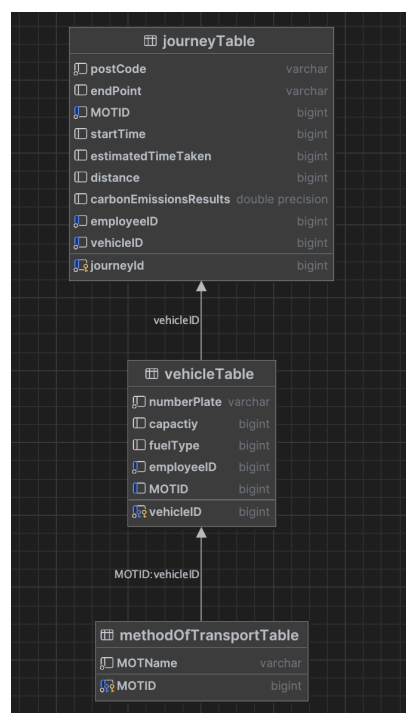
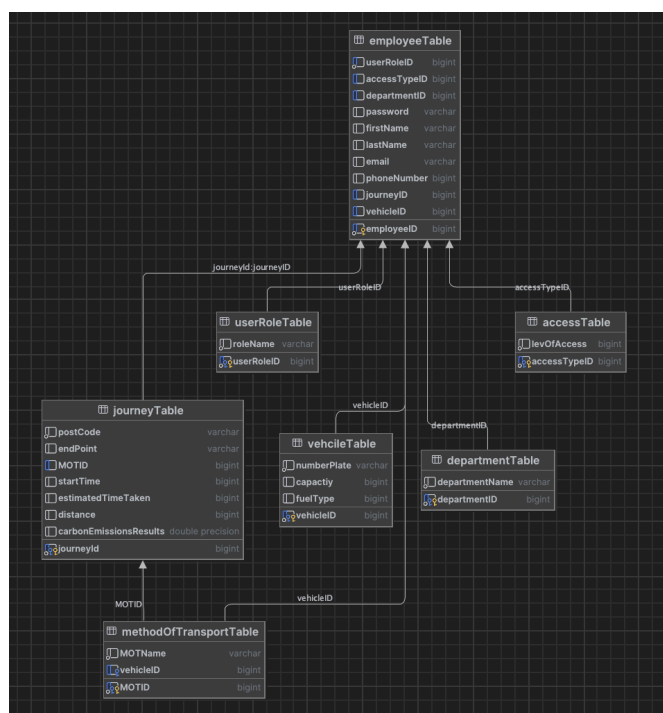
Database Seeding Doc

Firstly, to ensure employee registration doesn't rely on other tables, I'll outline a quick change to our schema.

Minor Change:

- Changed *vehcileTable* to *vehicleTable*.
- Changed *capactiy* to *capacity*.
- *levOfAccess* inside *accessTable* changed to *varchar*.

Change of Relation Dynamic between MOT, Journey and Vehicle Tables



The diagrams (left and right) show the changes made to the database- before and after, respectively.

The relations have been cleared up to allow the MOT table to stay relatively small, only as big as the number of different methods of transport.

Journey Table Start/End Points Customised

The journey table has also been updated to allow storage of latitude and longitude for both start and end points.

Post code has remained the same, although for added accuracy, I've included an address field, which should contain the street name and number, but not the post code, as there is a separate field for that.

journeyTable	
postCode	varchar
MOTID	bigint
estimatedTimeTaken	bigint
distance	bigint
carbonEmissionsResults	double precision
employeeID	bigint
vehicleID	bigint
address	varchar
startLat	bigint
startLong	bigint
endLat	bigint
endLong	bigint
journeyID	bigint

First steps in data seeding procedures

To allow the seeding of employees, we first must handle the dependencies on which the employee table relies on: access type, department, and user role type.

Access Type specifies if the employee is a driver or passenger.

User Role specifies if the user is an administrator, or if they just have normal user privileges.

Finally, department specifies which department each user belongs to.

We start with the different user roles as follows.

```
INSERT INTO "userRoleTable" VALUES (0, 'Driver');  
INSERT INTO "userRoleTable" VALUES (1, 'Passenger');
```

For now, the access table has been seeded with these values for each access type.

```
INSERT INTO "accessTable" VALUES (0, 'Employee');  
INSERT INTO "accessTable" VALUES (1, 'Admin');
```

Initial departments have been allocated here.

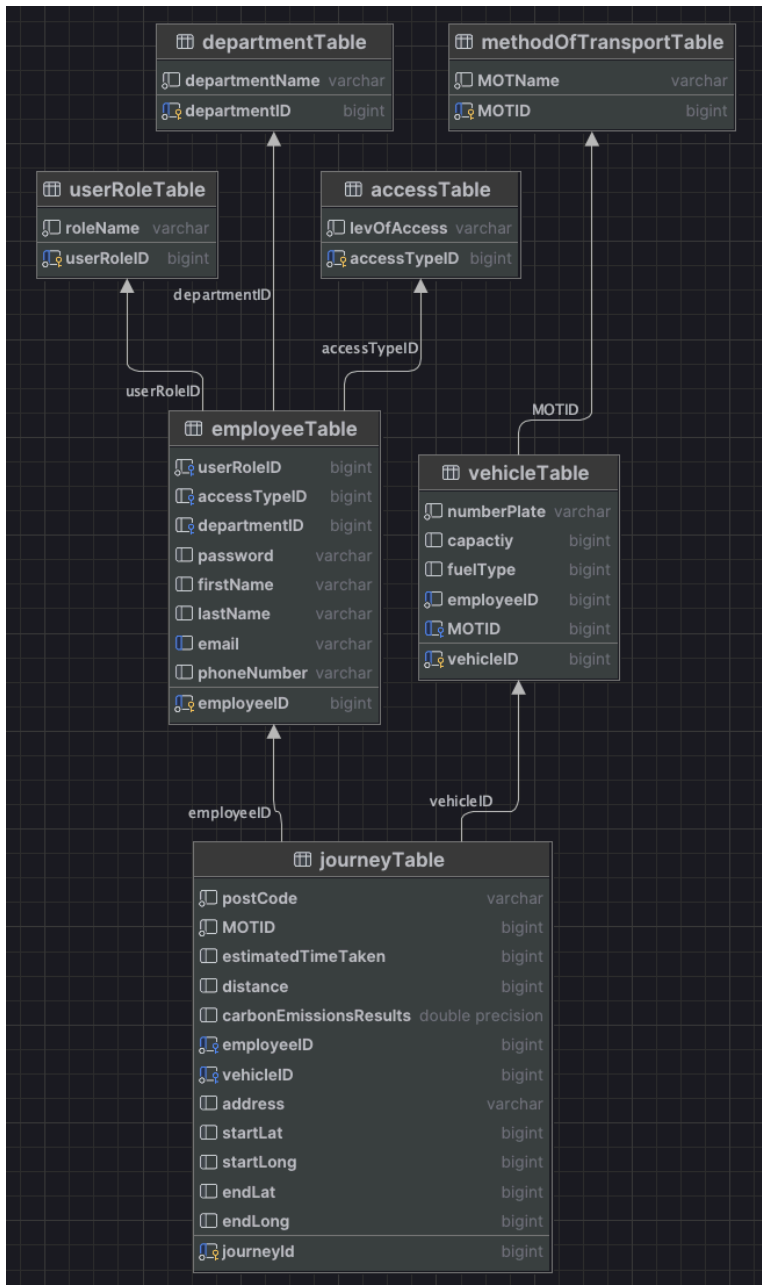
```
INSERT INTO "departmentTable" VALUES (0, 'HR');  
INSERT INTO "departmentTable" VALUES (1, 'Marketing');  
INSERT INTO "departmentTable" VALUES (2, 'IT');  
INSERT INTO "departmentTable" VALUES (3, 'Sales');  
INSERT INTO "departmentTable" VALUES (4, 'Management');
```

Seeding Employees Table

The next step was to seed the employees table, which has been done with a python script, utilising the Faker library to generate the data.

The script can be found on the Gitlab under the appropriate branch.

Database Foreign Key Change



There had to be a quick swap in all the foreign keys in the table, as initially they were the other way around. The above image shows the diagram after all changes had been made.

Vehicle and MOT Table Seeding

First we need to manage the dependencies of the journey table, the vehicle and MOT tables.

The basic MOT's should include transport by car, bus and walking. Others can be implemented later / edited from the app dashboard.

```
insert into "methodOfTransportTable" values (0,'Car');
insert into "methodOfTransportTable" values (1,'Bus');
insert into "methodOfTransportTable" values (2,'Walk');
```

Next is our vehicle table. Each Journey has to be completed by a specific vehicle.

- If an employee walks to work, although this doesn't exactly fit nicely into human language, we should have a dedicated 'vehicle id' for this, which would enable us to distinguish it from other vehicles.
- This could also be verified by looking at the method of transport for this row, which should correspond to walking.
- Similarly implementation for biking and others will be done.

Assumptions

- For fuel type, I've assumed that 0 is diesel and 1 is petrol, but the definitions can be changed.
- Also for now I've assumed all vehicles inserted are cars.

Journey Table Seeding

To seed the journeys table, I've had to leave a few fields as NULL, as I haven't had direct access to all the data. However, this is for the sake of minimising waiting times between the sub-teams, and all the rows can be all modified later.

The fields left as NULL are as follows: `estimatedTimeTaken`, `distance`, `carbonEmissionsResults`.

The full script for the journey table seeding can be found on the [Gitlab](#).